

LESSON PLAN	COURSE: MHF4U
Date: Lessons #4	Unit of Study: Chapter #2 Polynomial Equations and Inequalities
Materials / Resources: - Advanced Function 12 Study Guide - Chapter #2 Formula, Aid and Homework Sheet(s) - Graphing Technology, e.g. desmos.com - KWL Chart	Lesson Focus - Families of Polynomial Functions - Solve Inequalities Using Technology - Solve Factorable Polynomial Inequalities Algebraically
Prior Knowledge: - Polynomials that shares the same roots - Inequalities of a function and its meaning	
Overall Expectations: 1. Solve problems involving polynomial and simple rational equations graphically and algebraically 2. Demonstrate an understanding of solving polynomial and simple rational inequalities	
Specific Expectations: 1.8 Determine the equation of the family of polynomial functions with a given set of zeros and of the member of the family that passes through another given point [e.g., a family of polynomial functions of degree 3 with zeros 5, -3, and -2 is defined by the equation $f(x) = k(x - 5)(x + 3)(x + 2)$, where k is a real number, $k \neq 0$; the member of the family that passes through $(-1, 24)$ is $f(x) = -2(x - 5)(x + 3)(x + 2)$] 4.1 Explain, for polynomial and simple rational functions, the difference between the solution to an equation in one variable and the solution to an inequality in one variable, and demonstrate that given solutions satisfy an inequality (e.g., demonstrate numerically and graphically that the solution to $\frac{1}{x+1} < 5$ is $x < -1$ or $x > -\frac{4}{5}$) 4.2 Determine solutions to polynomial inequalities in one variable [e.g., solve $f(x) \geq 0$, where $f(x) = x^3 - x^2 + 3x - 9$] and to simple rational inequalities in one variable by graphing the corresponding functions, using graphing technology, and identifying intervals for which x satisfies the inequalities 4.3 Solve linear inequalities and factorable polynomial inequalities in one variable (e.g., $x^3 - x^2 > 0$) in a variety of ways (e.g., by determining intervals using x-intercepts and evaluating the corresponding function for a single x-value within each interval; by factoring the polynomial and identifying the conditions for which the product satisfies the inequality), and represent the solutions on a number line or	

algebraically (e.g., for the inequality $x^4 - 5x^2 + 4 < 0$, the solution represented algebraically is $-2 < x < -1$ or $1 < x < 2$)

Learning Goals:

- Students will be able to produce the equations of a family of polynomial functions given certain conditions
- Students will demonstrate their understanding of inequality by applying on a number line
- Students will understand the difference between the solutions of a polynomial equation and a polynomial inequality
- Students will know how to solve polynomial inequalities algebraically and while using technology

Success Criteria:

1. By the end of this lesson I will be able to determine the equation of a family of polynomial functions that satisfy given conditions
2. By the end of this lesson I will be able to solve linear and factorable polynomial inequalities and represent the solutions on a number line
3. By the end of this lesson I will be able to explain the difference between the solution to a polynomial equation and a polynomial inequality
4. By the end of this lesson I will be able to solve polynomial inequalities algebraically and using technology

Assessment Strategies:

Formative: homework for chapters 2.4, 2.5 & 2.6

Summative: test #1

Time:

Part 1

Introduction:

Take up / check homework from lesson #1 and check for student's understanding. If needed, go over specific questions.

Time:

Part 2

Lesson:

Chapter 2.4 Families of Polynomial Functions

Using technology such as desmo.com to graph multiple different polynomials that contain the same roots and observe these equations. Determine through investigation how equations of polynomials are families with one another and how that changes with the equations (e.g., a family of polynomial equations with zeros $a_1, a_2, \dots, a_n$ can be represented by an equation of the form $y = k(x - a_1)(x - a_2)\dots(x - a_n)$ where $k \in \mathbb{R}, k \neq 0$)

Chapter 2.5 Solve Inequalities Using Technology

Using technology such as desmo.com, graph a polynomial equation and then show that $f(x) < 0$ is the equivalent to the points of the graph that is below the x axes and

	$f(x) > 0$ is the equivalent to the points of the graph that is above the x axes and thus the corresponding domain is the range of x where $f(x) < 0$ and $f(x) > 0$ are satisfied. Apply these restrictions onto a number line. <u>Chapter 2.6 Solve Factorable Polynomial Inequalities Algebraically</u> Using numbers lines as a guide, solve for factored polynomials by graphing the roots and then the equation to determine the domain that satisfies the inequality. Pay close attention to the sign of the leading coefficient of the equation as that complete changes the satisfied range of domain. Also give examples where the roots have orders other than 1 (e.g. $0 < x(x - 2)^2(x - 1)^3$)
Time: Part 3	Wrap-up: Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions. Use KWL chart to better understand student's learning needs for chapter #2.
Notes, reminders & homework: <u>Reminder:</u> - Quiz #2 (to be written before / after current class) - Test #2 (to be written before / after lesson #8) <u>Homework:</u> Chapter 2.4: # 1 – 9, 12 Chapter 2.5: # 1 – 8, 10, 12 Chapter 2.6: # 1 – 9, 11 Chapter #2: Challenge Questions (preparation for test #2) C1, C8, C9	
Lesson Reflection:	

